

Lab 04: NetLogo application: fire-spread lab

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Part A: Build the model

```
globals [  
  initial-trees  
  burned-trees  
]  
  
to setup  
  clear-all  
  ask patches [  
    if random 100 < density [ set pcolor green ]  
  ]  
  ask patches with [ pxcor = min-pxcor ] [  
    if pcolor = green [ set pcolor red ]  
  ]  
  set initial-trees count patches with [ pcolor = green or pcolor = red ]  
  set burned-trees 0  
  reset-ticks  
end  
  
to go  
  if not any? patches with [ pcolor = red ] [ stop ]  
  ask patches with [ pcolor = red ] [  
    ask neighbors4 with [ pcolor = green ] [  
      set pcolor red  
    ]  
    set pcolor grey  
    set burned-trees burned-trees + 1  
  ]  
end
```

```
tick
end
```

Task A1. Embed your screenshot of the final state of the world at density = 60:

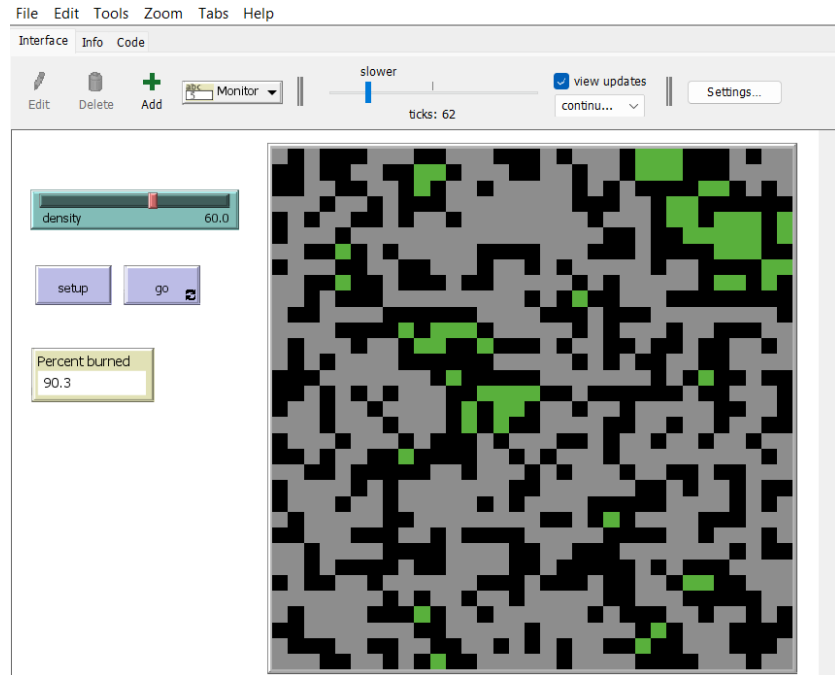


Figure 1: Fire Spread NetLogo model with 60% Density

Task A2. Percent burned at the end of the run: (*value from the monitor*)

Percent Burned at end of run: 90.3%

Part B: Find the critical density

Task B1. Complete the table: ten runs at each density, percent burned each time.

Den- sity	Run 1	Run 2	Run 3	Run 4	Run 5	Run 6	Run 7	Run 8	Run 9	Run 10	Aver- age
40	8.4%	10.4%	7.2%	12.8%	7.5%	16.0%	17.3%	19.4%	16.0%	8.8%	12.4%
50	20.1%	45.4%	39.7%	13.3%	33.1%	60.2%	43.9%	40.8%	50.6%	24.3%	37.1%
55	76.0%	88.7%	75.2%	75.6%	48.9%	61.9%	21.7%	40.1%	31.2%	60.2%	58.0%
58	85.9%	86.2%	64.2%	94.8%	68.1%	85.8%	74.0%	66.0%	91.4%	85.9%	80.2%

Density	Run 1	Run 2	Run 3	Run 4	Run 5	Run 6	Run 7	Run 8	Run 9	Run 10	Average
59	72.1%	93.4%	91.6%	84.2%	74.3%	93.3%	94.1%	58.3%	86.4%	83.8%	83.2%
60	85.8%	89.7%	93.1%	89.1%	92.7%	94.9%	88.4%	91.2%	79.0%	72.2%	87.6%
62	89.2%	84.1%	84.9%	84.5%	79.5%	94.6%	96.7%	74.2%	84.9%	92.0%	86.5%
65	96.7%	96.1%	90.6%	91.3%	96.5%	96.5%	97.8%	95.0%	99.1%	97.5%	95.7%
70	98.6%	97.4%	97.9%	99.6%	96.7%	98.0%	98.4%	98.9%	99.1%	98.3%	98.3%
80	100.0%	99.8%	100.0%	99.8%	100.0%	100.0%	99.9%	99.9%	100.0%	99.5%	99.9%

Task B2. Plot mean percent burned vs density.

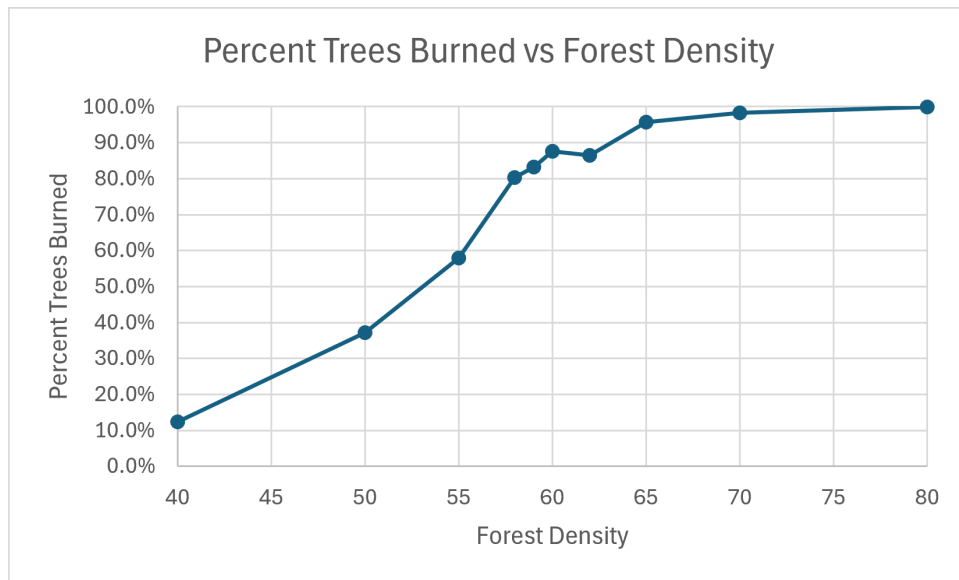


Figure 2: Plot of Trees Burned compared with the density of the forest.

Task B3 (critical-density estimate).

The percent jumps from “small” at 55% density to “large” at 58% density - meaning the critical density is around 58%.

Task B4 (variability near the threshold).

Variability is lower near the lowest and highest densities - at low densities, the chance of a tree existing in one of the 4 neighbouring spots is always low, almost always causing the fire to fizzle out - at high densities, the chance of there being any isolated trees at all, conversely, is low. At the threshold, the trees near the starting locations of the fire will randomly either have really strong pathways or very weak ones which are purely dependent on the random setup of the forest, while low and high density will have almost none or total pathways respectively.

Part C: Summary ODD documentation

Replace each <...> placeholder with your own text about the fire model. Keep the **bold** phrases as they are and keep ODD's key words in *italics*. Aim for one short paragraph per element.

The overall *purpose* of our model is to illustrate the relationship between forest density and the ability of fire to spread and destroy a given forest. **Specifically, we are addressing the following questions:** How does the density of trees in a forest affect the ability of a forest fire to spread, and what is the threshold density for which the destruction will increase the most? **To consider our model realistic enough for its purpose, we use the following *patterns*:** having a distinct density threshold at around 59% that shows a sudden jump in the forest percentage destroyed once you've passed it which will show up for each random forest layout.

The model includes the following *entities*: Patches that represent two global entities with pcolor: initial-trees and burned-trees. **The *state variables* characterising these entities are listed in Table 1. The *spatial* and *temporal* resolution and extent** are 33 patches by 33 patches (for a total of 1089 patches), where each patch represents the area of one large tree canopy, and where each tick represents one minute of time. The run ends when there are no more "red" (on fire) initial-trees, meaning the forest fire has burnt out.

The most important *processes* of the model, which are repeated every *time step*, are first, at the start of the loop, the model checks if the fire has burnt out (i.e. whether there are no red, on fire initial-trees anymore), and if so it stops - then, if the simulation doesn't stop, it checks whether any green (healthy) trees are in the four cardinal directions of an 'on fire' tree - if so, those trees are themselves set on fire. Finally, the initial-tree patch that was originally on fire is then converted to a burnt-tree patch and turned grey, and adds 1 to the burnt tree count.

The most important *design concepts* of the model are the emergence of a critical density threshold of trees resulting in a sudden jump in forest destruction only from spread by individual trees, the randomness/stochasticity of the tree placement - making the model repeatable in order to gain a more accurate insight - and finally, the source of the fire coming from a "fire front" rather than a single ignition point (therefore representing a patch of forest in a larger forest fire rather being the start of its own).

The model is initialized with a random set of green initial-trees generated based on the input density - where the initial-trees at the far left column are set to red, on fire trees, 'igniting' the forest fire. The initial trees value is set to count the number of these green or red patches, and burned-trees is initialised as 0 (as none should start out in the simulation). The slider starts at 60%, with an increment of 1%.

Table 1. Entities and state variables of the model.

Entity	Variable	Description	Possible values	Units
Patch	pcolor	The colour of each patch, representing initial trees/on fire trees and burnt trees	green, red and grey	none
Global	density	the percentage of patches that start as initial trees (green and red)	0-100	%
Global	inital-tree	green or red tree representing a living tree in the forest	0-1089	patches
Global	burned-tree	grey, dead tree that has been burned	0-1089	patches